GHANA COMMUNICATION TECHNOLOGY UNIVERSITY

INSTITUTE OF CONTINUING AND DISTANCE EDUCATION (ICDE)

COURSE CODE: CSIS151

COURSE TITLE: FUNDAMENTALS OF COMPUTING

UNIT 1:

Session 1: **INTRODUCTION TO COMPUTERS**

Session 2: Introduction to Computers Characteristics of

Session 3: Computers Evolution of Computers Generations of

Session 4: Computers Classification of Computers Computer

Session 5: Systems Concept Application of Computers

Session 6:

Session 7:

UNIT 1:

INTRODUCTION TO COMPUTERS

OVERVIEW

This unit provides a foundational understanding of computers, their evolution, and their role in modern society. Students will explore the basic concepts of computing, including the characteristics of computers, their historical development, and their classification. The unit also covers the components of a computer system and the various applications of computers in different fields. By the end of this unit, students will have a solid grasp of what computers are, how they have evolved, and how they are used in everyday life.

KEY TOPICS:

1. Introduction to Computers
2. Characteristics of Computers
3. Evolution of Computers
4. Generations of Computers
5. Classification of Computers
6. Computer Systems Concept
7. Applications of Computers

SESSION 1: INTRODUCTION TO COMPUTERS

OBJECTIVE

By the end of this session, students will be able to: define what a computer is, explain its purpose, and identify the basic components of a computer system (hardware and software). Students will also understand the role of computers in modern society and provide examples of how computers are used in various fields.

SHORT NOTES:

1.Write down issues that are not clear here; and

2.Difficult topics if any

1.1 Definition With Examples

1.1.1 What is a computer?

A computer is an electronic device that processes data and performs tasks according to a set of instructions called a program. Computers are used in almost every aspect of modern life, from education and business to entertainment and healthcare. They are capable of performing complex calculations, storing vast amounts of data, and communicating with other computers over networks. Additionally, it operates under the control of instructions (software) and performs four primary functions:

Input: Accepts data from users or devices.

Processing: Performs calculations or logical operations on data.

Storage: Retains data and instructions for future use.

Output: Produces results or information for users.

Figure 1.0 A Desktop Computer System

The merits or advantages of computers include the following.

- **Increased efficiency and productivity:** Computers can perform tasks much faster and more accurately than humans, allowing for increased efficiency and productivity in various industries.
- **Storage and organization of information:** Computers can store large amounts of data and organize it in a way that is easily accessible and searchable.
- **Improved communication:** Computers enable people to communicate easily and instantly with others, regardless of their location.
- **Access to information and resources:** The internet provides access to a vast amount of information and resources that would otherwise be difficult or impossible to obtain.
- **Automation of repetitive tasks:** Computers can automate repetitive and mundane tasks, freeing up time and resources for more important work.

The demerits or disadvantages of computers include the following.

- **Dependence on technology:** Over-reliance on computers can lead to problems if they break down or malfunction, leading to loss of productivity and data.
- **Security risks:** Computers can be vulnerable to viruses, malware, and hacking, leading to data breaches and other security risks.
- **Social isolation:** The overuse of computers can lead to social isolation and reduced face-to-face interaction, leading to social and emotional problems.
- **Environmental impact:** The production and disposal of computers can harm the environment due to the use of resources and the creation of electronic waste.
- **Job displacement:** Automation and the use of computers can lead to job displacement in certain industries, requiring workers to adapt to new skill sets or find new employment.

1.1.2 Key Points:

- **Definition**: A computer is a programmable machine that receives input, processes it, and produces output.
- **Purpose**: Computers are designed to automate tasks, solve problems, and enhance productivity.
- **Components**: A computer system consists of hardware (physical components) and software (programs and data).

Examples:

- **Personal Computers (PCs)**: Used for everyday tasks like browsing the internet, creating documents, and playing games.
- **Supercomputers**: Used for complex scientific calculations, such as weather forecasting and molecular modeling.
- **Embedded Systems**: Found in everyday devices like washing machines, cars, and smartphones.

Figure 1.1 A modern computer setup (desktop, laptop, or tablet)

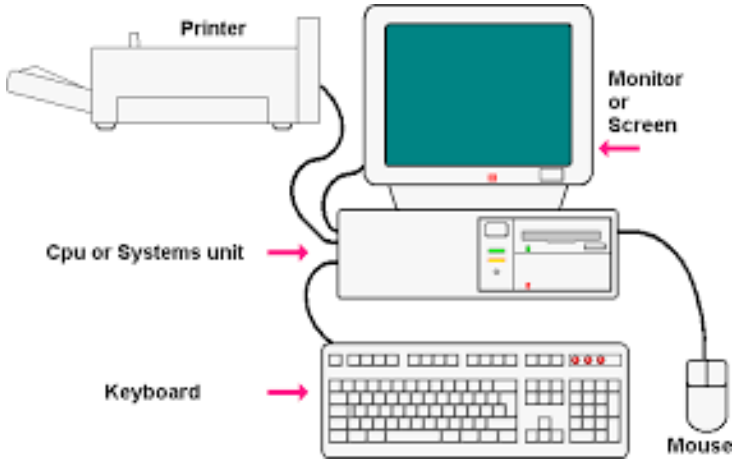

Figure 1.2 Diagram of Basic components of a computer (CPU, monitor, keyboard, mouse)

The following are the different components of a computer.

Central Processing Unit (CPU): The CPU is the brain of the computer. All the important decisions and manipulations are carried out by the CPU. CPU is further divided into the Arithmetic Logic Unit (ALU) and the Control Unit (CU).

Motherboard: The motherboard is the main circuit that connects all the components of a computer.

Memory: The RAM (Random Access Memory) is used for quick access to data.

Storage: The storage components include Hard Disk Drives (HDD) and Solid-State drives (SDD). They are used for storing data for the long term.

Input devices: The various devices via which we can input data into the computer from the user are called input devices. Examples of input devices include keyboard, mouse, and scanners.

Output devices: Output devices are the devices via which the computer gives out the output to the users. Examples of output devices include printers and speakers.

1.1.5 Self-Assessment Questions

Assignment 1: Briefly explain a computer and state and basic steps in data processing.

Assignment 2: Briefly distinguish between hardware and software.

Assignment 3: Write a short essay (300-500 words) on the role of computers in modern society. Include examples of how computers are used in different fields such as education, healthcare, and entertainment.

Assignment 4: Create a diagram or infographic showing the basic components of a computer system. Label each component and briefly describe its function.

SESSION 2: CHARACTERISTICS OF COMPUTERS

OBJECTIVE:

By the end of this session, students will be able to describe the key characteristics of computers, including speed, accuracy, storage, versatility, automation, and reliability. Students will also be able to explain how these characteristics enable computers to perform complex tasks and provide real-world examples of these characteristics in action.

SHORT NOTES:

1.Write down issues that are not clear here; and

2.Difficult topics if any

1.2.1 What are the Characteristics of a Computer?

Computers have several key characteristics that make them powerful and versatile tools. These include speed, accuracy, storage capacity, versatility, automation, and reliability.

1.Speed:

- **Definition**: Computers can perform millions or even billions of calculations per second. This speed is measured in **hertz (Hz)**, with modern processors operating in **gigahertz (GHz)**.
- **Explanation**: The speed of a computer allows it to process large amounts of data in a very short time. For example, a supercomputer can perform trillions of calculations per second, making it ideal for tasks like weather forecasting and scientific simulations.
- **Example**: A modern computer can perform complex mathematical calculations in milliseconds, whereas a human would take significantly longer to perform the same task.

2.Accuracy:

- **Definition**: Computers perform tasks with a high degree of precision, provided the input data and instructions are correct.
- **Explanation**: Computers are designed to follow instructions exactly as programmed. Errors in output are usually due to incorrect input or programming mistakes, not the computer itself.
- **Example**: Financial institutions rely on computers to perform accurate calculations for transactions, investments, and payroll processing. A single error in these calculations could lead to significant financial losses.

3.Storage:

- **Definition**: Computers can store vast amounts of data in various forms, such as text, images, videos, and programs.
- **Explanation**: Storage capacity is measured in **bytes**, with modern computers capable of storing terabytes (TB) or even petabytes (PB) of data. This allows users to save and retrieve information as needed.
- **Example**: Cloud storage services like Google Drive and Dropbox allow users to store and access large amounts of data from anywhere in the world.

4.Versatility:

- **Definition**: Computers can perform a wide range of tasks, from simple calculations to complex simulations and multimedia processing.
- **Explanation**: Versatility is achieved through software, which allows computers to be programmed for different tasks. A single computer can be used for word processing, gaming, video editing, and more.

- **Example**: A personal computer can be used for browsing the internet, creating documents, playing games, and editing videos, all with the right software.

5.Automation:

- **Definition**: Computers can perform tasks automatically once they are programmed to do so.
- **Explanation**: Automation reduces the need for human intervention in repetitive or complex tasks. For example, automated systems can control manufacturing processes, manage inventory, and even drive cars.
- **Example**: Automated teller machines (ATMs) allow customers to perform banking transactions without the need for a human teller.

6.Reliability:

- **Definition**: Modern computers are highly reliable and can operate for long periods without failure.
- **Explanation**: Reliability is achieved through robust hardware design and error-checking mechanisms in software. Computers are designed to handle errors and recover from failures, ensuring consistent performance.
- **Example**: Servers in data centers are designed to operate 24/7 without interruption, ensuring that online services like email and social media are always available.

1.2.2 Real-World Applications:

Speed: Supercomputers are used for scientific research, such as simulating nuclear reactions or predicting climate change.

- **Accuracy**: Computers are used in medical imaging to provide precise diagnoses based on X-rays, MRIs, and CT scans.

6.Reliability:

- **Definition**: Modern computers are highly reliable and can operate for long periods without failure.
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1.2.2 Real-World Applications:

Speed: Supercomputers are used for scientific research, such as simulating nuclear reactions or predicting climate change.

- **Accuracy**: Computers are used in medical imaging to provide precise diagnoses based on X-rays, MRIs, and CT scans.

- **Storage**: Cloud storage services allow businesses to store and access large amounts of data remotely.
- **Versatility**: Personal computers are used for a wide range of tasks, from creating documents to editing videos.
- **Automation**: Automated systems are used in manufacturing to assemble products with high precision.
- **Reliability**: Online banking systems rely on computers to process transactions accurately and securely.

Self-Assessment Questions

Assignment 1: What is the importance of speed in computers?

Assignment 2: Describe how computers achieve accuracy.

Assignment 3: Provide an example of computer versatility.

Assignment 4: What is scalability, and why is it important?

Assignment 5: How does connectivity enhance computer

utility?

SESSION 3:

EVOLUTION OF

COMPUTERS

OBJECTIVE:

By the end of this session, students will be able to trace the historical development of computers from early mechanical devices to modern electronic systems. Students will identify key milestones in computer history, such as the invention of the abacus, mechanical calculators, and early electronic computers like ENIAC and UNIVAC.

SHORT NOTES:

1.Write down issues that are not clear here; and

2.Difficult topics if any

1.3.1 Overview of the Computer's Evolution:

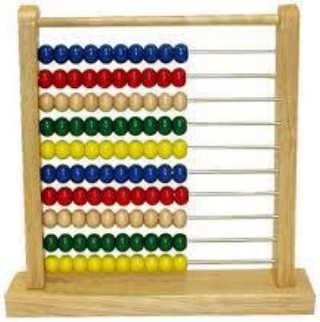
The evolution of computers has been marked by significant advancements in technology, from early mechanical devices to modern electronic computers. The history of computers can be divided into several key periods, including the development of the abacus, mechanical calculators, and electronic computers.

1.Early Computing

Devices: Abacus:

- **Definition**: The abacus is one of the earliest known computing devices, used for basic arithmetic calculations.
- **Explanation**: The abacus consists of beads arranged on rods, which can be moved to perform addition, subtraction, multiplication, and division. It was widely used in ancient civilizations, including China, Greece, and Rome.
- **Example**: Merchants in ancient China used the abacus to calculate trade transactions.

Figure 1.3 An Abacus



Mechanical Calculators:

- **Definition**: Mechanical calculators were early devices designed to perform arithmetic operations automatically.
- **Explanation**: These devices used gears, levers, and wheels to perform calculations. They were the precursors to modern electronic calculators.
- **Examples**:
  - **Pascaline**: Invented by Blaise Pascal in the 17th century, it could perform addition and subtraction.
  - **Leibniz Wheel**: Invented by Gottfried Wilhelm Leibniz, it could perform multiplication and division.

Figure 1.4 Mechanical Calculators

2. Early Electronic Computers:

ENIAC (Electronic Numerical Integrator and Computer):

- **Definition**: ENIAC was one of the first general-purpose electronic computers, developed in the 1940s.
- **Explanation**: ENIAC used vacuum tubes to perform calculations and was programmed using switches and cables. It was used for complex calculations, such as ballistic trajectories during World War II.
- **Example**: ENIAC could perform 5,000 additions per second, a significant achievement at the time.

Figure 1.5: ENIAC

UNIVAC (Universal Automatic Computer):

- **Definition**: UNIVAC was the first commercial computer, introduced in the 1950s.
- **Explanation**: UNIVAC used magnetic tape for data storage and was used for business and scientific applications. It marked the beginning of the computer age in the private sector.
- **Example**: UNIVAC was used to predict the outcome of the 1952 U.S. presidential election.

Figure 1.6: UNIVAC

3.The Rise of Modern Computers:

Transistors and Integrated Circuits:

- **Definition**: The invention of transistors in the 1940s and integrated circuits (ICs) in the 1950s revolutionized computing.
- **Explanation**: Transistors replaced vacuum tubes, making computers smaller, faster, and more reliable. Integrated circuits further miniaturized computers by combining multiple transistors on a single chip.
- **Example**: The IBM 1401, a transistor-based computer, was widely used in businesses in the 1960s.

Figure 1.7: Transistors and Integrated Circuits.

Microprocessors:

- **Definition**: Microprocessors, introduced in the 1970s, are the "brains" of modern computers.
- **Explanation**: A microprocessor integrates the functions of a computer's central processing unit (CPU) onto a single chip. This innovation led to the development of personal computers (PCs).
- **Example**: The Intel 4004, released in 1971, was the first commercially available microprocessor.

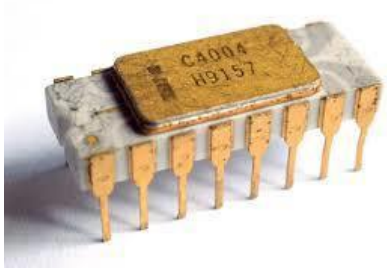


Figure 1.8: Early Microprocessors (Intel 4004)

4.The Personal Computer Revolution:

Apple II and IBM PC:

- **Definition**: The Apple II (1977) and IBM PC (1981) were among the first successful personal computers.
- **Explanation**: These computers brought computing power to homes and small businesses, making computers accessible to the general public.
- **Example**: The Apple II was popular for gaming and educational software, while the IBM PC became the standard for business computing.

Figure 1.9: Apple II and IBM PC

Graphical User Interface (GUI):

- **Definition**: The introduction of the graphical user interface (GUI) in the 1980s made computers more user-friendly.
- **Explanation**: GUIs replaced text-based interfaces with visual elements like icons, windows, and menus, making it easier for non-technical users to interact with computers.
- **Example**: The Apple Macintosh (1984) was one of the first computers to feature a GUI.

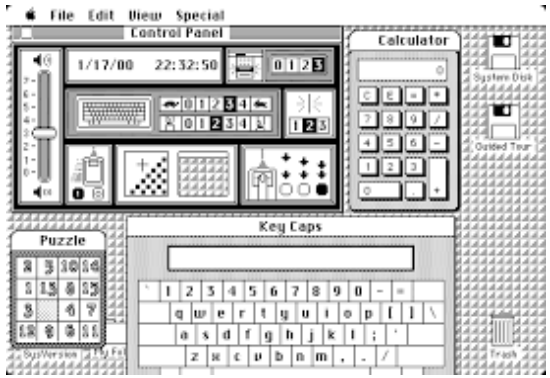


Figure 1.10: Early GUIs like the Apple Macintosh

5.The Internet and Beyond:

The Internet:

- **Definition**: The development of the internet in the late 20th century transformed how computers are used.
- **Explanation**: The internet allows computers to communicate and share information globally, enabling applications like email, the World Wide Web, and online shopping.
- **Example**: The first web browser, Mosaic, was released in 1993, making the internet accessible to the general public.

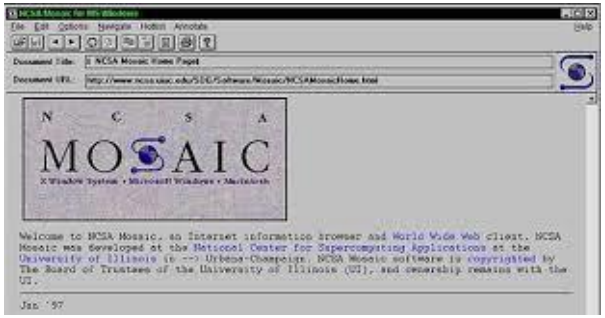


Figure 1.11: Early Web Browsers (Mosaic).

Modern Computing:

- **Definition**: Today's computers are faster, smaller, and more powerful than ever before.
- **Explanation**: Modern computers use advanced technologies like multi-core processors, solid-state drives (SSDs), and cloud computing. They are used in a wide range of applications, from artificial intelligence to virtual reality.
- **Example**: Smartphones, which are essentially pocket-sized computers, have become an essential part of daily life.

Figure 1.12: Modern Computers

1.3.2 Real-World Applications:

- **Early Computing Devices**: The abacus is still used in some parts of the world for teaching basic arithmetic.
- **Early Electronic Computers**: ENIAC and UNIVAC paved the way for modern computing by demonstrating the potential of electronic computers.
- **Modern Computers**: Personal computers and smartphones have revolutionized how we work, communicate, and access information.

1.3.3 Self-Assessment Questions

1. **Assignment 1**: Create a timeline of the evolution of computers, starting from the abacus to modern-day computers. Include key milestones and technological advancements.
    - **Example**: Include the invention of the abacus, mechanical calculators, ENIAC, transistors, microprocessors, and the internet.
2. **Assignment 2**: Write a research paper (800-1000 words) on the impact of early electronic computers (like ENIAC and UNIVAC) on the development of modern computing. Discuss how these early machines paved the way for today's technology.

- **Example**: Discuss how ENIAC's ability to perform complex calculations influenced the development of scientific computing.

SESSION 4: EVOLUTION OF COMPUTERS

OBJECTIVE

By the end of this session, students will be able to identify and describe the five generations of computers, from vacuum tubes to artificial intelligence. Students will understand the technological advancements that define each generation and provide examples of computers from each generation.

SHORT NOTES

1.Write down issues that are not clear here; and

2.Difficult topics if any

1. Overview of the Computer's Evolution:

The evolution of computers has been marked by significant advancements in technology, from early mechanical devices to modern electronic computers. The history of computers can be divided into several key periods, including the development of the abacus, mechanical calculators, and electronic computers.

- **First Generation (1940s-1950s):** Used vacuum tubes, which were large and consumed a lot of power. Vacuum tubes were expensive and produced a lot of heat, which made these computers very expensive and only affordable to large organizations. Machine language was the programming language used for these computers, and they could not multitask. The ENIAC was the first electronic general-purpose computer that used 18,000 vacuum tubes and was built in 1943 for war-related calculations. Examples of the first generation include EDVAC, IBM-650, IBM-701, Manchester Mark 1, Mark 2, etc.

Figure 1.13: First Generation Computer

- **Second Generation (1950s-1960s)**: Used transistors, which were smaller and more reliable. The second generation of computers was developed in the late 1950s and 1960s. These computers replaced vacuum tubes with transistors making them smaller, faster, and more efficient. This was done as transistors were more reliable than vacuum tubes, required less maintenance, and generated less heat. Second-generation computers were smaller and more portable, making them accessible to a wider audience. Magnetic core memory was also introduced in this generation, which was faster and more reliable. This laid the foundation for further developments, paving the way for the third generation that used integrated circuits.

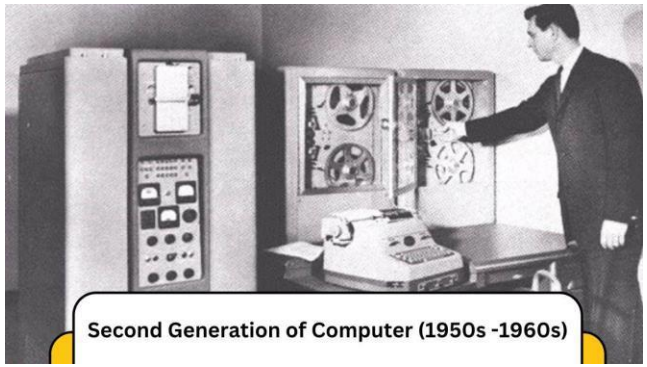


Figure 1.14: Second Generation Computer

- **Third Generation (1960s-1970s)**: Used integrated circuits (ICs), which further reduced size and cost. This generation used microchips or integrated circuits, making it possible to create smaller, cheaper, and much faster computers. The third

generation of computers was much faster than previous generations, with computational times reduced from microseconds to nanoseconds. New input devices like the mouse and keyboard were introduced, replacing older methods like punch cards. New functionalities, like multiprogramming time-sharing, and remote processing, were introduced, allowing for more efficient use of computer resources.

Figure 1.15: Third Generation Computer

- **Fourth Generation (1970s-Present)**: Use microprocessors, leading to personal computers. The fourth-generation computers were developed in 1972 after the third generation that used microprocessors. They used Very Large Scale Integrated (VLSI) circuits, which contained about 5000 transistors capable of performing complex activities and computations. These computers

were more adaptable, had more primary storage capacity, were faster and more reliable than previous generations, and were also portable, small, and required less electricity. This generation had the first supercomputers, used complex programming languages like C, C++, DBASE, etc., and could perform many accurate calculations.

Figure 1.16: Fourth Generation Computer

- **Fifth Generation (Present and Beyond)**: Focus on artificial intelligence and quantum computing. The fifth generation of computers emerged after the fourth generation and is still being developed. Computers of the fifth generation use artificial intelligence (AI) to perform various tasks. These computers use programming languages such as Python, R, C#, Java, etc., as input methods. These computers can perform intricate tasks such as image recognition, human speech interpretation, natural language understanding, etc. Examples of the fifth generation include laptops, desktops, notebooks, Chromebooks, etc.

Figure 1.17: Fifth Generation Computer

2. Examples:

- **First Generation**: ENIAC and UNIVAC were early examples of first-generation computers.
- **Second Generation**: IBM 1401 and IBM 7094 were popular second-generation computers.
- **Third Generation**: IBM System/360 and DEC PDP-8 were key third-generation computers.
- **Fourth Generation**: Personal computers like the Apple II and IBM PC revolutionized computing.
- **Fifth Generation**: Modern AI-driven systems like IBM Watson and Google DeepMind.

Computer Generations	Switching Device Used	Storage Devices Used	Operating Systems Used	Programming Languages
FIRST	Vaccum Tube	Magnetic Drum	Batch Operating System	Machine Language (bits)
SECOND	Transistor	Magnetic Core	MultiBatch Operating System	Assembly Language
THIRD	Integrated Circuits (IC)	Magnetic Core	Real Time Sharing Operating Systems	FORTRAN, COBOL etc.
FOURTH	Large Scale Integrated Circuits (LSI)	Semiconductor Memory,Winchestor Disk	Time sharing Operating Systems,MS DOS,UNIX	Advanced FORTRAN-77,PASCAL,ADA etc.
FIFTH	Very LargeScale IC (VLSI)	Optical Disk	Windows,LINUX,GUI-LINUX,Android	HighLevel Languages C,JAVA etc.

1.4.3

Figure 1.17: Comparison of the Five Generations of Computers Self-Assessment Questions

Assignment 1: What was the main technology used in the first generation of computers?

Assignment 2: Name one key milestone in the evolution of computers.

Assignment 3: How did transistors improve computing?

Assignment 4: What defines fifth-generation computing?

Assignment 5: Explain how integrated circuits revolutionized computer design.

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SESSION 5: CLASSIFICATION OF COMPUTERS

OBJECTIVE

By the end of this session, students will be able to classify computers based on their size, processing power, and intended use. Students will identify and describe the main categories of computers, including supercomputers, mainframes, minicomputers, and microcomputers, and provide examples of each type.

SHORT NOTES

1.Write down issues that are not clear here; and

2.Difficult topics if any

1.5.1 Classifying Computers

Classification of computers is particularized into three main categories, which include;

1. Classification of Computers based on Size and Capacity.
2. Classification of Computer based on Type.
3. Classification of Computer according to Purpose.

Classification of Computers based on Size and Capacity.

The above classification of computers is based on the internal memory and speed of processing. The computer is classified into four parts based on size and capability including;

1. Micro Computer
2. Mini Computer
3. Mainframe computer
4. Super Computer

Micro Computer

The microcomputer is an electronic device with a microprocessor as its central processing unit (CPU). Microcomputer was formerly a commonly used term for personal computers, particularly any of a class of small digital computers whose CPU is contained on a single integrated semiconductor chip. Examples include calculators, mobile phones, notebooks, and workstations. A personal computer is a computer that can perform all of its input, processing, output, and storage activities by itself. A personal computer contains a processor, memory, and one or more input, output, and storage devices. Personal computers also often contain a communications device.

Mini Computer

A minicomputer is a computer that is smaller, less expensive, and less powerful than a mainframe or supercomputer but more expensive and more powerful than a personal computer. Minicomputers are used for scientific and engineering computations, business transaction processing, file handling, and database management. Examples include Apple iPad, Samsung tab, Dell Inspiron, HP mini- 110, Netbook, iPod, Apple iPhone, and Samsung Galaxy series.

Mainframe Computer

A mainframe is a large, expensive, powerful computer that can handle hundreds or thousands of connected users simultaneously. Mainframes store tremendous amounts of data, instructions, and information. Most major corporations use mainframes for business activities. With mainframes, enterprises can bill millions of customers, prepare payroll for thousands of employees, and manage thousands of items in inventory. Mainframes also can act as servers in a network environment. Servers and other mainframes can access data and information from a mainframe. People also can access programs on the mainframe using terminals or personal computers.

Uses of Mainframe Computer

- Performing ATM cash withdrawals and deposits. During the process, communication between the mainframe and remote computer will help accomplish the financial transactions at hand.
- Business transactions that use credit cards or pre-paid cards.
- Online electronic transactions.
- Cloud storage.
- Handling of patient records in major hospitals.

Super Computer

A supercomputer is the fastest, most powerful computer, and the most expensive. The fastest supercomputers are capable of processing more than one quadrillion instructions in a single second. With weights that exceed 100 tons, these computers can store more than 20,000 times the data and information of an average desktop computer. Applications requiring complex, sophisticated mathematical calculations use supercomputers. Large-scale simulations and applications in medicine, aerospace, automotive design, online banking, weather forecasting, nuclear energy research, and petroleum exploration use a supercomputer.

Figure 1.18: Super Computer

Classification of computers according to type is given below.

The computer is classified into three parts based on Type basis Hardware Design and Data Handling. You can also call it the classification of computers by functionality. These computers different roles in different fields. These include:

1. Analog Computer

2. Digital Computer

3. Hybrid Computer

Analog Computer

An analog computer is a device that performs tasks using continuous data. Analog devices are used to measure voltage, pressure, electric current, and temperature. These quantities vary continuously while measuring the quantity, such as the temperature of a person which changes consistently. Examples include Thermometers, speedometers, voltmeters, etc.

Digital Computer

A digital computer is a machine that stores data in a numerical format and performs operations on that data using mathematical manipulation. The output of digital devices is in the form of binary numbers. The best examples of digital devices are desktops, laptops, smartphones, and calculators.

Analog Computer *Digital Computer*

Figure 1.19: Analog and Digital Computer

Hybrid Computer

A hybrid computer is a type of computer that offers the functionalities of both a digital and an analog computer. It is designed to include a working analog unit that is powerful for calculations, yet has a readily available digital memory. In large industries and businesses, a hybrid computer can be used to incorporate logical operations as well as provide efficient processing of differential equations. **Examples** include Gasoline stations, Electrocardiogram machines, Ultrasound machines, etc.

Classification of Computer According to Purpose

The computer is classified into two parts According to Purpose. Every computer is designed with a different design and different purpose. Every computer is designed for a particular place and some specific purpose

1. Special Purpose computer

2. General-purpose computer

Special Purpose Computer

Special purpose computer: This is a computer designed for a particular type of application only. It is defined as a stored program digital computer, whose architecture is oriented towards one or more applications. Examples include; computers meant for process control in an industry, computers meant for air traffic control, computers used in robots, and computers used for desktop publishing.

Figure 1.19: Special Purpose Computers

General Purpose Computer

A general-purpose computer is a computer designed to solve a wide variety of problems from different fields. The hardware and software of such systems are adaptable to different environments. Examples: Most of the computers used in our normal day-to-day work are general-purpose computers.

Functions of Computers

A computer performs five major operations or functions irrespective of its size and make. These are

1.it accepts data or instructions as input,

2.it stores data and instruction

3.it processes data as per the instructions,

4.4. it controls all operations inside a computer, and

SESSION 6: COMPUTER SYSTEMS CONCEPT

OBJECTIVE

By the end of this session, students will be able to explain the components of a computer system, including hardware, software, data, and users. Students will understand how these components interact to process data and perform tasks, and they will be able to create a diagram illustrating the architecture of a computer system.

SHORT NOTES

1.Write down issues that are not clear here; and

2.Difficult topics if any

1.6.1 What are Computer Systems Concepts?

A computer system is a complex integration of hardware, software, data, and users that work together to perform tasks. Understanding the components of a computer system and how they interact is essential for grasping how computers function. In this session, we will explore the four main components of a computer system: hardware, software, data, and users. We will also discuss how these components work together to process information and perform tasks.

1.Hardware:

Definition:

Hardware refers to the physical components of a computer system that can be touched and seen. These components are responsible for input, processing, output, and storage.

Key Components:

- **Central Processing Unit (CPU)**: The "brain" of the computer that performs calculations and executes instructions.
- **Memory (RAM)**: Temporary storage that holds data and instructions while the computer is running.
- **Storage Devices**: Permanent storage for data and programs (e.g., hard drives, SSDs).
- **Input Devices**: Devices that allow users to input data into the computer (e.g., keyboard, mouse).
- **Output Devices**: Devices that display or output data from the computer (e.g., monitor, printer).

Examples:

- **CPU**: Intel Core i7 or AMD Ryzen processors.
- **Memory**: 8GB or 16GB RAM modules.
- **Storage**: 1TB hard drive or 512GB SSD.
- **Input Devices**: Keyboard, mouse, scanner.
- **Output Devices**: Monitor, printer, speakers.

1.6.1 What are Computer Systems Concepts?

A computer system is a complex integration of hardware, software, data, and users that work together to perform tasks. Understanding the components of a computer system and how they interact is essential for grasping how computers function. In this session, we will explore the four main components of a computer system: hardware, software, data, and users. We will also discuss how these components work together to process information and perform tasks.

1.Hardware:

Definition:

Hardware refers to the physical components of a computer system that can be touched and seen. These components are responsible for input, processing, output, and storage.

Key Components:

- **Central Processing Unit (CPU)**: The "brain" of the computer that performs calculations and executes instructions.
- **Memory (RAM)**: Temporary storage that holds data and instructions while the computer is running.
- **Storage Devices**: Permanent storage for data and programs (e.g., hard drives, SSDs).
- **Input Devices**: Devices that allow users to input data into the computer (e.g., keyboard, mouse).
- **Output Devices**: Devices that display or output data from the computer (e.g., monitor, printer).

Examples:

- **CPU**: Intel Core i7 or AMD Ryzen processors.
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- **Output Devices**: Monitor, printer, speakers.

2.Software:

Definition:

Software refers to the programs and operating systems that control the hardware and allow users to perform specific tasks. Without software, hardware cannot function.

Types of Software:

- **System Software**: Manages the hardware and provides a platform for running application software (e.g., operating systems like Windows, macOS, Linux).
- **Application Software**: Programs designed to perform specific tasks for users (e.g., Microsoft Word, Google Chrome, Adobe Photoshop).

Examples:

- **System Software**: Windows 10, macOS Catalina, Linux Ubuntu.
- **Application Software**: Microsoft Office Suite, Google Chrome, Adobe Creative Cloud.

3.Data:

Definition:

Data refers to the raw information that is processed by the computer. It can take many forms, including text, numbers, images, audio, and video.

Types of Data:

- **Text**: Documents, emails, web pages.

- **Numbers**: Spreadsheets, financial data.
- **Images**: Photos, graphics.
- **Audio**: Music, voice recordings.
- **Video**: Movies, video clips.

Examples:

- **Text**: A Word document containing an essay.
- **Numbers**: An Excel spreadsheet with financial data.
- **Images**: A JPEG photo of a landscape.
- **Audio**: An MP3 file of a song.
- **Video**: An MP4 file of a movie.

4.Users:
Definition:

Users are the people who interact with the computer system to perform tasks. They input data, use software, and interpret the output produced by the computer.

Types of Users:

- **End Users**: Individuals who use the computer for personal or professional tasks (e.g., students, professionals, gamers).
- **System Administrators**: Professionals who manage and maintain computer systems (e.g., IT staff in organizations).
- **Examples:**
- **End Users**: A student using a laptop to write an essay.
- **System Administrators**: An IT professional managing a company's network.

1.6.2 How Components Work Together

Input → Processing → Output:

- **Input**: Users input data into the computer using input devices (e.g., typing on a keyboard).
- **Processing**: The CPU processes the data according to instructions provided by the software.
- **Output**: The processed data is displayed or output using output devices (e.g., displaying text on a monitor).

Example:

- **Input**: A user types a document in Microsoft Word.
- **Processing**: The CPU processes the text and stores it in memory.
- **Output**: The document is displayed on the monitor and can be printed using a printer.

1.6.3 Real-World Applications

1. **Hardware**: In a business setting, computers with powerful CPUs and large storage capacities are used for data analysis and financial modeling.
2. **Software**: Operating systems like Windows and macOS are used to manage hardware resources, while application software like Microsoft Excel is used for data analysis.
3. **Data**: In healthcare, patient data (e.g., medical records, and test results) is stored and processed by computers to assist in diagnosis and treatment.
4. **Users**: Students use computers to complete assignments, while IT professionals manage and maintain computer systems in organizations.

1.6.4. SELF-ASSESSMENT QUESTIONS

1. Define a computer system.
2. What are the components of a computer system?
3. How do input and output devices interact?
4. Explain the role of storage devices in information processing.
5. Name two examples of integrated computer systems.

SESSION 7:

APPLICATION OF

COMPUTERS

OBJECTIVE

By the end of this session, students will be able to discuss the various applications of computers in different fields, such as education, healthcare, business, entertainment, and communication. Students will provide real-world examples of how computers are used in each field and explain the impact of computers on modern society.

SHORT NOTES

1.Write down issues that are not clear here; and

2.Difficult topics if any

1.1.2 Applications of Computers

Computers have become an integral part of modern life, revolutionizing the way we work, learn, communicate, and entertain ourselves. In this session, we will explore the various applications of computers in different fields, including education, healthcare, business, entertainment, and communication. By understanding these applications, students will gain insight into how computers have transformed society and how they continue to shape the future.

1. Education:

Role of Computers in Education:

Computers have transformed the field of education by providing new ways to learn, teach, and access information. They enable personalized learning, online education, and access to vast amounts of educational resources.

Examples:

- **Online Learning Platforms**: Websites like Coursera, Khan Academy, and Udemy offer online courses on a wide range of subjects.
- **Educational Software**: Programs like Microsoft Office and Adobe Creative Suite are used to teach skills like word processing, graphic design, and data analysis.
- **Virtual Classrooms**: Tools like Zoom and Google Classroom allow students and teachers to interact in real-time, even when they are not physically in the same location.

2. Healthcare:

Role of Computers in Healthcare:

Computers play a critical role in healthcare by improving diagnosis, treatment, and patient care. They are used for medical imaging, electronic health records (EHR), and research.

Examples:

- **Medical Imaging**: Computers are used to process images from X-rays, MRIs, and CT scans, helping doctors diagnose conditions more accurately.
- **Electronic Health Records (EHR)**: EHR systems store patient information digitally, making it easier for healthcare providers to access and share medical records.
- **Research**: Computers are used to analyze large datasets in medical research, such as genomics and drug discovery.

3. Business:

Role of Computers in Business:

Computers have revolutionized the way businesses operate by automating tasks, improving efficiency, and enabling global communication. They are used for accounting, data analysis, customer relationship management (CRM), and more.

Examples:

- **Accounting**: Software like QuickBooks and SAP is used to manage financial transactions, payroll, and taxes.
- **Data Analysis**: Tools like Microsoft Excel and Tableau are used to analyze business data and make informed decisions.
- **Customer Relationship Management (CRM)**: Systems like Salesforce help businesses manage customer interactions and improve customer service.

Entertainment:

Role of Computers in Entertainment:

Computers have transformed the entertainment industry by enabling the creation and distribution of digital content. They are used for gaming, streaming, and digital art.

Examples:

- **Gaming**: Computers and gaming consoles like PlayStation and Xbox are used to play video games, which have become a major form of entertainment.
- **Streaming**: Platforms like Netflix, YouTube, and Spotify allow users to stream movies, TV shows, and music.
- **Digital Art**: Software like Adobe Photoshop and Blender is used to create digital art, animations, and visual effects.

5. Communication:

Role of Computers in Communication:

Computers have revolutionized communication by enabling instant communication across the globe. They are used for email, social media, and video conferencing.

Examples:

- **Email**: Services like Gmail and Outlook allow users to send and receive emails instantly.
- **Social media**: Platforms like Facebook, Twitter, and Instagram enable users to share information and connect with others.